

# Robust Dialect Classification for Hindi Speakers

## Using Augmented Speech Data and Transformer-Based Models

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### Abstract

This report presents the work completed during the semester for the undergraduate project on robust dialect classification. The initial objective was to train a dialect classifier on a corpus where the same sentence was spoken across multiple dialects, so that the variation captured by the model would be primarily dialectal rather than lexical. Since such a controlled dataset was not available for this project, the problem was reformulated in a practical and well-motivated way using speaker metadata.

The NISP corpus was selected because it contains multilingual speech along with rich speaker information, including native language and native region. From this corpus, speakers whose native language was Hindi were selected, and both their Hindi and English recordings were retained. The speaker's native state was then used as a proxy label for regional dialect variation. The resulting dataset was highly imbalanced, so a two-stage balancing strategy was designed: majority classes were under-sampled to a target threshold of 350 utterances, and minority classes were augmented using Vocal Tract Length Perturbation (VTLP).

In addition, because the recordings had substantial duration variation, a spectral-flux based salient segment extraction procedure was used to identify a more informative 3-second region from each utterance instead of depending only on conventional padding. To test robustness, a noisy version of the dataset was generated by adding Gaussian noise at random SNR levels between 10 dB and 30 dB, and the same noisy corpus was reused across model evaluations.

The prepared data was evaluated using Dasheng, Wav2Vec2, and Whisper-based classifiers. On the clean dataset, the complete proposed pipeline improved Dasheng accuracy from 20.73 % with padding-only input to 88.85 %. On the noisy dataset, the same pipeline improved Dasheng accuracy from 18.30 % to 77.79 %, showing that the data preparation strategy contributes substantially to robustness. Wav2Vec2 also showed a small improvement on the clean dataset, from 98.98 % to 99.07 %.

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# 1 Introduction

Automatic dialect classification is an important problem in speech processing because dialectal variation affects pronunciation, prosody, phone realization, and speaking style. These differences influence the behaviour of downstream systems such as automatic speech recognition, speaker profiling, multilingual assistants, and forensic speech analysis. At the same time, dialect classification is difficult because dialects of the same language often share vocabulary and grammar, and the differences between them are usually more subtle than the differences between separate languages.

The original aim of this project was to perform dialect classification in a controlled setting where the same textual content is spoken by speakers from different dialect backgrounds. Such a setting is ideal because it reduces lexical variation and allows the model to focus more directly on dialect-related acoustic cues. However, after searching for suitable resources, such a dataset was not found for the present work. Instead of stopping at that point, the project adopted a realistic alternative: use a dataset with rich speaker metadata and construct regional dialect proxy labels using the native state of the speaker.

The NISP dataset was selected for this purpose because it provides speech recordings together with metadata such as native language, regional background, age, gender, and body-build characteristics [1]. In this project, the selection criterion was speakers whose native language is Hindi. For those speakers, both Hindi and English recordings were retained, and the speaker’s `Native_State` field was used as the target label. This transformed the task from direct dialect recognition into metadata-guided regional dialect proxy classification. Although this is not equivalent to using linguistically verified dialect labels, it remains a defensible and useful setting for studying robust regional speech classification.

Once the dataset was prepared, three major difficulties became clear. First, the state-wise class distribution was extremely skewed. Second, the utterances had highly variable durations. Third, a robustness study required the pipeline to be evaluated not only on clean data but also under noisy conditions. The main work of the semester therefore focused on building a robust end-to-end data preparation and evaluation pipeline that addresses all three issues.

# 2 Related Work

Previous work in dialect identification has progressed through several stages. Early systems relied heavily on carefully engineered acoustic, prosodic, and phonotactic features. For Hindi, Rao and Koolagudi explored spectral and prosodic features for five Hindi dialect conditions and reported dialect identification using AANN and SVM-based classifiers [2]. Similarly, Sinha and colleagues studied Hindi dialect identification using MFCC, PLP, MF-PLP, and prosodic cues together with auto-associative neural networks, showing the continued importance of handcrafted acoustic-phonetic representations in the early literature [3]. In Arabic dialect identification, Biadisy and Hirschberg demonstrated

that prosodic information can significantly improve performance over purely phonotactic baselines, which showed that rhythm and intonation carry useful dialect cues [4].

A later shift toward stronger utterance-level representations can be seen in the work of Ali et al., who combined phonetic and lexical features with bottleneck-feature i-vectors and multi-class SVMs for Arabic broadcast speech [5]. More recent work has moved toward end-to-end and transfer-learning approaches. Wang et al. showed that a dialect identification system can benefit from transfer learning from a multilingual ASR encoder [6]. Recent studies on low-resource Indian dialect identification also report that pretrained representations such as wav2vec 2.0, especially when paired with transformer-based backends, outperform traditional MFCC-based pipelines by a clear margin [9, 8]. At the same time, robustness remains a major challenge: Sullivan et al. showed that even strong self-supervised Arabic dialect identification systems can be vulnerable to domain shift, which is highly relevant for real-world deployment [7].

Taken together, the literature suggests three clear trends: a shift from handcrafted features to learned representations, increasing use of transfer learning and self-supervised speech models, and continued concern about data scarcity, imbalance, and robustness. The present project is aligned with these trends, but focuses specifically on building a practical, data-centric pipeline for Hindi regional dialect proxy classification.

## 3 Methodology

### 3.1 Problem Formulation and Dataset Selection

The project began with the search for a dialect dataset containing the same sentence spoken across different dialects. Since this was not available, the task was reformulated using regional metadata. The working assumption was that a speaker’s native state can act as a reasonable proxy for dialect or accent variation for speakers whose native language is Hindi.

The NISP dataset was chosen because it was explicitly created for speaker profiling and contains both speech recordings and metadata. The NISP paper reports a total of 345 speakers across five Indian languages and English, recorded using a consistent microphone setup in quiet indoor environments [1]. For this project, speakers whose native language is Hindi were selected. The paper reports that this Hindi-native speaker subset contains 103 speakers, including 76 male and 27 female speakers [1]. Both Hindi and English utterances spoken by these Hindi-native speakers were retained in the derived dataset.

**Table 1:** Summary of the source corpus and the derived project dataset.

| Item                        | Value                                                                   |
|-----------------------------|-------------------------------------------------------------------------|
| Source corpus               | NISP: NITK-IISc Multilingual Multi-accent Speaker Profiling dataset     |
| Languages in NISP           | Hindi, Kannada, Malayalam, Telugu, Tamil, and English                   |
| Total speakers in NISP      | 345                                                                     |
| Hindi-native speakers used  | 103                                                                     |
| Utterance languages used    | Both Hindi and English recordings of the selected Hindi-native speakers |
| Number of state labels      | 19                                                                      |
| Utterances before balancing | 7,839                                                                   |
| Utterances after balancing  | 6,460                                                                   |

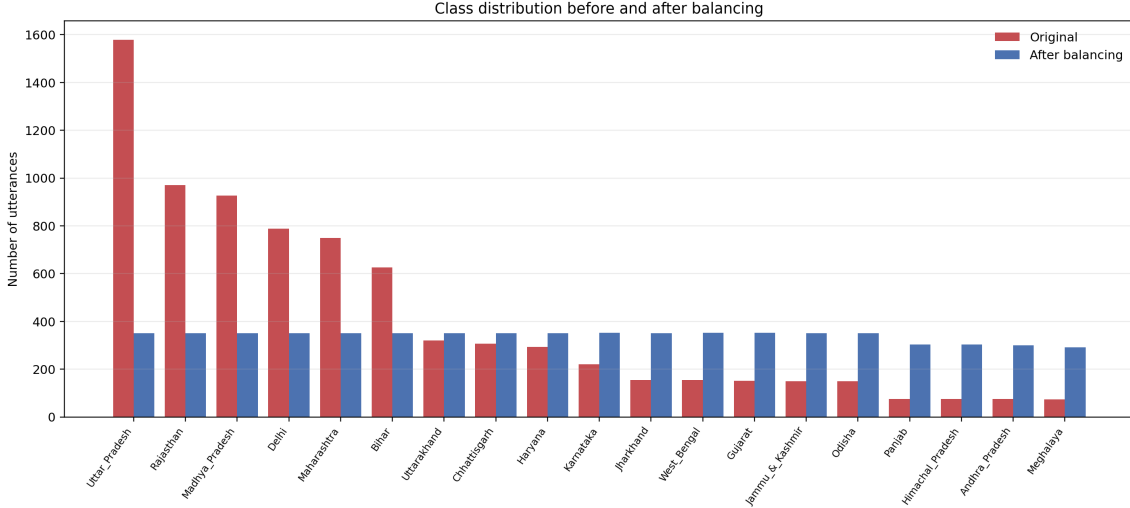
### 3.2 Metadata-Guided Label Construction

The audio filenames were parsed to recover speaker IDs, and these IDs were then aligned with the speaker metadata table. Only speakers whose native language was Hindi were retained. For these speakers, both Hindi and English recordings were kept in the working dataset. From the merged table, the `Native_State` field was used as the target label. This produced a 19-class regional dataset containing speech from Bihar, Delhi, Madhya Pradesh, Maharashtra, Rajasthan, Uttar Pradesh, Uttarakhand, Chhattisgarh, Haryana, Karnataka, Jharkhand, West Bengal, Gujarat, Jammu & Kashmir, Odisha, Panjab, Himachal Pradesh, Andhra Pradesh, and Meghalaya.

This design choice should be stated clearly: the labels used in this work are regional proxy labels, not expert-annotated linguistic dialect labels. The goal of the project was therefore not to claim perfect linguistic dialect recognition, but to construct and evaluate a robust regional speech classification pipeline that captures meaningful dialect-related variation.

### 3.3 Imbalance Analysis

The resulting dataset was strongly imbalanced. Uttar Pradesh contained 1,579 utterances, while Meghalaya contained only 73. The mean class size was 412.58, the median class size was 220, and the max/min ratio was approximately 21.63:1. Such an imbalance can strongly bias a classifier toward the largest classes and produce misleadingly optimistic performance on majority regions.



**Figure 1:** Class distribution before and after the balancing pipeline.

### 3.4 Balancing Strategy

To reduce imbalance, a target threshold of 350 utterances per state was chosen. This threshold was selected as a practical compromise. It is lower than the overall mean class size, which helps suppress the dominance of the largest states, but still high enough to avoid discarding too much data from the moderately sized classes. It also keeps the amount of required augmentation within a manageable range for the smaller classes.

The balancing process was performed in two stages:

1. **Under-sampling of majority classes:** all classes with more than 350 utterances were randomly reduced to 350 examples.
2. **Over-sampling of minority classes:** classes below 350 examples were augmented using VTLP.

**Table 2:** State-wise distribution before and after balancing.

| State            | Original    | After Under-sampling | After VTLP  |
|------------------|-------------|----------------------|-------------|
| Uttar Pradesh    | 1579        | 350                  | 350         |
| Rajasthan        | 971         | 350                  | 350         |
| Madhya Pradesh   | 926         | 350                  | 350         |
| Delhi            | 788         | 350                  | 350         |
| Maharashtra      | 749         | 350                  | 350         |
| Bihar            | 626         | 350                  | 350         |
| Uttarakhand      | 320         | 320                  | 350         |
| Chhattisgarh     | 306         | 306                  | 351         |
| Haryana          | 294         | 294                  | 351         |
| Karnataka        | 220         | 220                  | 352         |
| Jharkhand        | 155         | 155                  | 350         |
| West Bengal      | 154         | 154                  | 352         |
| Gujarat          | 151         | 151                  | 352         |
| Jammu & Kashmir  | 150         | 150                  | 351         |
| Odisha           | 150         | 150                  | 351         |
| Panjab           | 76          | 76                   | 304         |
| Himachal Pradesh | 76          | 76                   | 304         |
| Andhra Pradesh   | 75          | 75                   | 300         |
| Meghalaya        | 73          | 73                   | 292         |
| <b>Total</b>     | <b>7839</b> | <b>4300</b>          | <b>6460</b> |

### 3.5 VTLP-Based Augmentation

Vocal Tract Length Perturbation (VTLP) was used for minority-class augmentation. VTLP applies a frequency-axis warping that changes speaker-dependent spectral characteristics while preserving the utterance label. Jaitly and Hinton proposed VTLP as a speech augmentation strategy and showed that random perturbation of the frequency axis can improve recognition performance in data-limited settings [10].

In this project, the implementation used the following settings:

- sampling rate: 16 kHz,
- warp factor range: 0.9 to 1.1,
- high-frequency boundary: 4800 Hz,
- three augmented versions per source utterance,
- peak normalisation before saving the augmented signal.

The choice of the 0.9–1.1 range is consistent with the original VTLP motivation of introducing useful variation without creating unrealistic distortions [10]. After augmentation, the class counts became much more uniform, although the smallest classes still remained below 350 due to the limited number of original utterances available for expansion.

### 3.6 Salient Segment Extraction

The audio recordings showed large duration variation, with durations ranging from approximately 1.78 s to 42.34 s. Simply padding or truncating all recordings to a common length can be suboptimal because it may preserve silence, weakly informative segments, or irrelevant material. To address this issue, the project used a data-driven 3-second extraction strategy.

The exact algorithm used in the notebook is a **spectral-flux peak-picking algorithm over the STFT magnitude spectrum**. First, the short-time Fourier transform is computed with `n_fft = 2048` and `hop_length = 512`, and the magnitude spectrogram is obtained. Then spectral flux is computed as the frame-wise change between consecutive magnitude spectra:

$$\text{Flux}(t) = \sqrt{\sum_f (S(f, t) - S(f, t - 1))^2}, \quad (1)$$

where  $S(f, t)$  denotes the magnitude spectrum at frequency bin  $f$  and frame  $t$ . The resulting flux contour is passed to a peak-picking stage using `scipy.signal.find_peaks`, and only peaks above the mean flux level are considered. These peaks are sorted by strength, and the strongest peak is selected as the centre of the most informative speech region. A 3-second window is then extracted around this peak. If the end of the window goes beyond the signal boundary, the window is shifted so that a full segment is obtained whenever possible. If a segment is still shorter than the required model input length after resampling, the signal is repeated from the beginning until the target duration is reached. This avoids large zero-padded silent regions.

### 3.7 Noisy Dataset Construction

To assess robustness, a noisy counterpart of the dataset was created. This noisy corpus was generated once and then reused while testing the different downstream models.

The noisy-data generation process was as follows:

1. load the clean waveform,
2. compute the RMS energy of the clean signal,
3. sample an SNR value uniformly from 10 to 30 dB,
4. compute the required noise RMS using

$$\text{RMS}_{\text{noise}} = \frac{\text{RMS}_{\text{signal}}}{10^{\text{SNR}/20}}, \quad (2)$$

5. generate zero-mean Gaussian noise with the required RMS,
6. add the noise to the clean signal,
7. peak-normalise the final waveform,

8. save the corrupted waveform and preserve the original class label.

The notebook shows that all 7,839 input files were processed and the resulting metadata was saved as `noisy_dataset.csv`. After this step, the same balancing and salient-segment extraction pipeline used for the clean dataset was applied again to the noisy dataset. This ensures that the noisy-data experiments measure the robustness of the full pipeline and not just the robustness of the classifier alone.

### 3.8 Model Evaluation

The processed dataset was evaluated using three pretrained models:

- **Dasheng base** (mispeech/dasheng-base): a masked audio encoder designed for general audio classification [11],
- **Wav2Vec2 XLS-R 300M** (facebook/wav2vec2-xls-r-300m) [12],
- **Whisper small** (openai/whisper-small) [13].

The Dasheng notebook uses an 80:20 train/validation split, while the Wav2Vec2 and Whisper notebooks use their own held-out evaluation splits defined in the respective experiments. The main runs were trained for 10 epochs. The reported metrics are accuracy, weighted F1-score, precision, and recall.

**Table 3:** Summary of the proposed experimental pipeline.

| Stage                  | Description                                                                                            |
|------------------------|--------------------------------------------------------------------------------------------------------|
| Data selection         | Hindi-native speakers selected from NISP; both Hindi and English recordings of those speakers retained |
| Label construction     | <b>Native_State</b> used as a regional dialect proxy label                                             |
| Imbalance handling     | Under-sampling to 350, followed by VTLP-based augmentation for minority classes                        |
| Duration normalisation | 3-second salient-segment extraction using spectral flux                                                |
| Short-signal handling  | Tiling/repetition instead of large zero-padding                                                        |
| Robustness test        | Gaussian noise added at random SNR between 10 and 30 dB                                                |
| Models evaluated       | Dasheng, Wav2Vec2 XLS-R 300M, Whisper small                                                            |



## 4 Results

### 4.1 Performance on the Clean Dataset

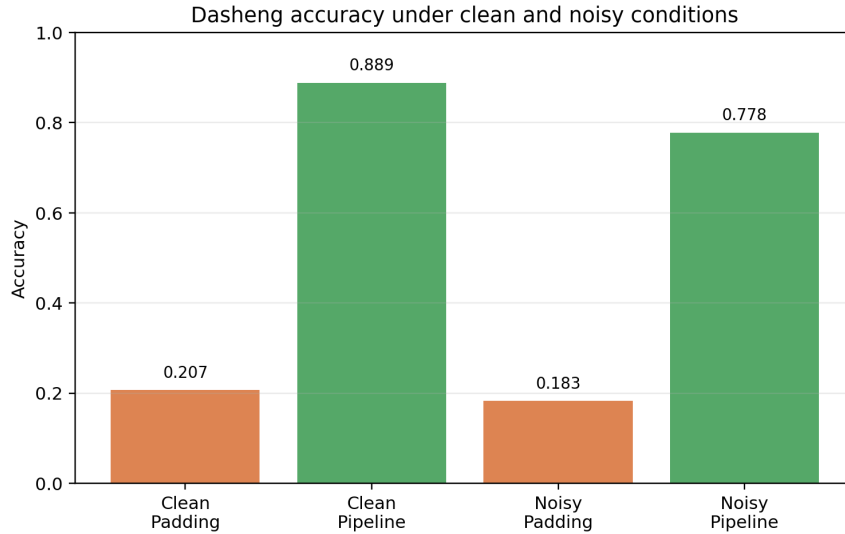
**Table 4:** Performance on the clean dataset.

| Configuration                           | Accuracy | F1     | Precision | Recall |
|-----------------------------------------|----------|--------|-----------|--------|
| Dasheng + Padding Only (baseline)       | 0.2073   | 0.1035 | 0.0697    | 0.2073 |
| Dasheng + VTLP Only                     | 0.8344   | 0.8148 | 0.7982    | 0.8344 |
| Dasheng + VTLP + Salient 3s Extraction  | 0.8885   | 0.8892 | 0.8911    | 0.8885 |
| Wav2Vec2 + Padding Only (baseline)      | 0.9898   | 0.9898 | 0.9899    | 0.9898 |
| Wav2Vec2 + VTLP + Salient 3s Extraction | 0.9907   | 0.9907 | 0.9909    | 0.9907 |

### 4.2 Performance on the Noisy Dataset

**Table 5:** Performance on the noisy dataset.

| Configuration                           | Accuracy | F1     | Precision | Recall |
|-----------------------------------------|----------|--------|-----------|--------|
| Dasheng + Padding Only                  | 0.1830   | 0.0927 | 0.0623    | 0.1830 |
| Dasheng + VTLP + Salient 3s Extraction  | 0.7779   | 0.7803 | 0.7849    | 0.7779 |
| Wav2Vec2 + Padding Only                 | 0.9739   | 0.9739 | 0.9741    | 0.9739 |
| Wav2Vec2 + VTLP + Salient 3s Extraction | 0.9834   | 0.9756 | 0.9776    | 0.9801 |
| Whisper + Padding Only                  | 0.9847   | 0.9848 | 0.9852    | 0.9847 |



**Figure 2:** Dasheng accuracy under clean and noisy conditions for the padding baseline and the proposed pipeline.

Across the noisy-data experiments, the proposed pipeline improved Dasheng from 18.30 % to 77.79 % accuracy and improved Wav2Vec2 from 97.39 % to 98.34 %. This indicates that the preprocessing pipeline contributes both to strong robustness recovery for weaker baselines and to smaller but still measurable gains for stronger pretrained models.

## 5 Discussion

The experimental results show that the strongest effect of the proposed pipeline appears in the Dasheng experiments. On clean data, padding-only input produces very poor performance, with accuracy slightly above 20 %. Introducing balancing through VTLP alone already raises Dasheng accuracy sharply to 83.44 %, which indicates that class imbalance was a major limiting factor in the original dataset. When salient 3-second extraction is added, the accuracy rises further to 88.85 %, suggesting that selecting an informative region of speech is more effective than presenting the model with arbitrary padded content.

The robustness experiments strengthen this conclusion. On the noisy dataset, Dasheng accuracy falls to 18.30 % for the padding-only baseline, but recovers to 77.79 % once the proposed balancing and extraction strategy is used. This is a large absolute improvement and supports the argument that the robustness of the overall system depends not only on the backbone model but also on how the training data is prepared.

The Wav2Vec2 results tell a slightly different story. Even the padding-only baseline is extremely strong on the clean dataset, which suggests that a large self-supervised encoder already captures useful representations for this task. The proposed pipeline still provides a small gain on the clean set, increasing accuracy from 98.98 % to 99.07 %. In the noisy condition, the baseline remains strong at 97.39 %, but the proposed pipeline further improves the accuracy to 98.34 %. Although this gain is much smaller than the gain observed for Dasheng, it still shows that balancing and salient-segment selection remain useful even for a strong pretrained encoder.

The Whisper result on noisy data is also very strong under the padding-only setup, with 98.47 % accuracy. This indicates that large pretrained audio models can remain highly stable under moderate synthetic noise. However, the main contribution of the present project is not simply to compare pretrained backbones. Rather, it is to show that a carefully designed data preparation pipeline can substantially improve performance and robustness, especially for models that are more sensitive to data imbalance and uninformative temporal structure.

At the same time, an important limitation should be acknowledged. The target classes in this work are derived from native state metadata and not from direct linguistic annotation of dialect. Therefore, the system should be interpreted as a regional dialect proxy classifier, not as a definitive dialect recogniser in the strict linguistic sense. Even with that limitation, the project provides a practical and reproducible framework for studying regional speech variation in a low-resource setting.

## 6 Conclusion

This project addressed the problem of robust dialect classification from speech of Hindi-native speakers in a realistic low-resource setting. Because an ideal same-sentence dialect corpus was not available, the task was reformulated using metadata from the NISP dataset. Speakers whose native language was Hindi were selected, both their Hindi and English recordings were retained, and the data was grouped according to native state to produce a 19-class regional speech dataset.

The major technical contributions of the semester were the design of a balancing pipeline based on under-sampling and VTLP augmentation, the development of a salient 3-second extraction method based on spectral-flux peak picking, and the construction of a shared noisy dataset for robustness testing.

The results show that these design choices matter. In particular, the complete pipeline produced a large improvement for Dasheng on both clean and noisy data. The findings support the broader view in the literature that data quality, representation quality, and robustness-oriented preprocessing are all critical for dialect identification, especially under limited or imperfect labelling conditions.

## 7 Future Scope

There are several meaningful directions for future work:

- collecting or identifying a true parallel-sentence dialect corpus for Hindi
- replacing state-based proxy labels with linguistically curated dialect labels
- extending the comparison to include the extraction-and-balancing pipeline for Whisper under noisy conditions
- testing the pipeline on the Vaani dataset
- Try to explore the reasons for such high accuracy for Wav2-Vec2 and Whisper, and find out whether there is any form of data leakage.
- Test the pipeline by splitting based on speaker ID rather than audio.
- Try to use a different method for adding the noise in the dataset

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